

Exploratory Data Analysis — Presidential Polling and Results

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Introduction

The 2012 and 2016 polling data sets are aggregated survey data by HuffPost Pollster, consisting of numerous state-level pre-election polls conducted by many different polling organisations. Each pollster utilised differing methodologies with factors such as interview mode, sample size and population, and field period varying between organisations. Fundamentally, the data sets measure the percentage of respondents who reported support for different presidential candidates, such as Obama/Romney in 2012 and Trump/Clinton in 2016. This data was retrieved from HuffPost Pollster via the pollstR R package on 10 January 2017 as part of the rOpenGov project. The polling data sets `state_polls_2012.csv` and `state_polls_2016.csv` can be found in the `data/raw/` folder within the [GitHub Repository](#)

Below is a detailed explanation of each column.

Key Variables:

- `poll_info`: text identifying the contest e.g. 2016-washington-d-c-president-trump-vs-clinton
- Obama / Romney (2012): Reported support for candidate (% , major-party)
- Trump, Clinton, Johnson, McMullin (2016): Reported support for each candidate (%)
- Undecided: Reported undecided voters (%)
- Other: Reported support for other candidates (%)
- `poll_id`: Poll identifier
- `pollster`: Polling organisation, sometimes with sponsor/partner appended
- `start_date`, `end_date`: Field period (YYYY-MM-DD)
- `sample_subpopulation`: Likely Voters / Registered Voters / Adults

- `sample_size`: Reported sample size
- `mode`: Interview mode (Live Phone, Internet, Mixed...)
- `partisanship`: Whether the poll is Nonpartisan, or partisan by Pollster or Sponsor
- `partisan_affiliation`: Party affiliation where partisan (Dem/Rep/None)

Data Types:

- Numerical: Obama, Romney, Trump, Clinton, Johnson, McMullin, Undecided, Other, `sample_size`
- Categorical: `poll_info`, `pollster`, `sample_subpopulation`, `mode`, `partisanship`, `partisan_affiliation`
- Temporal: `start_date`, `end_date`
- ID: `poll_id`

Varying factors such as sample size, sample population, field period, and interview mode means that observations are not based on identical polling methods or populations and thus, may not be directly comparable.

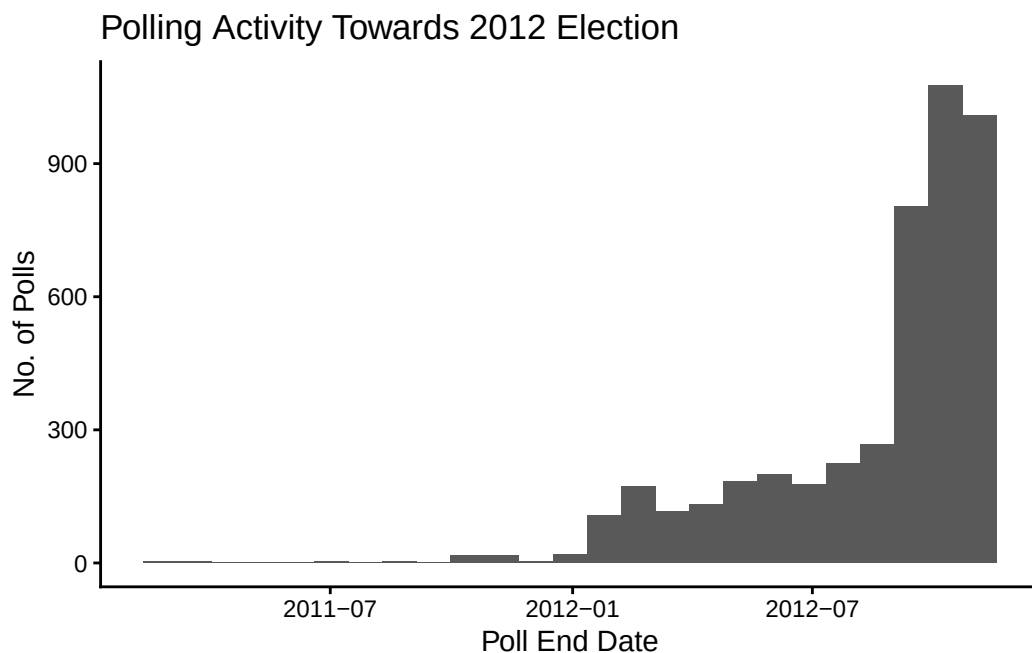


Figure 1: Polling coverage increased closer to Election Day in 2012, with more polls conducted in the last weeks of election campaign.

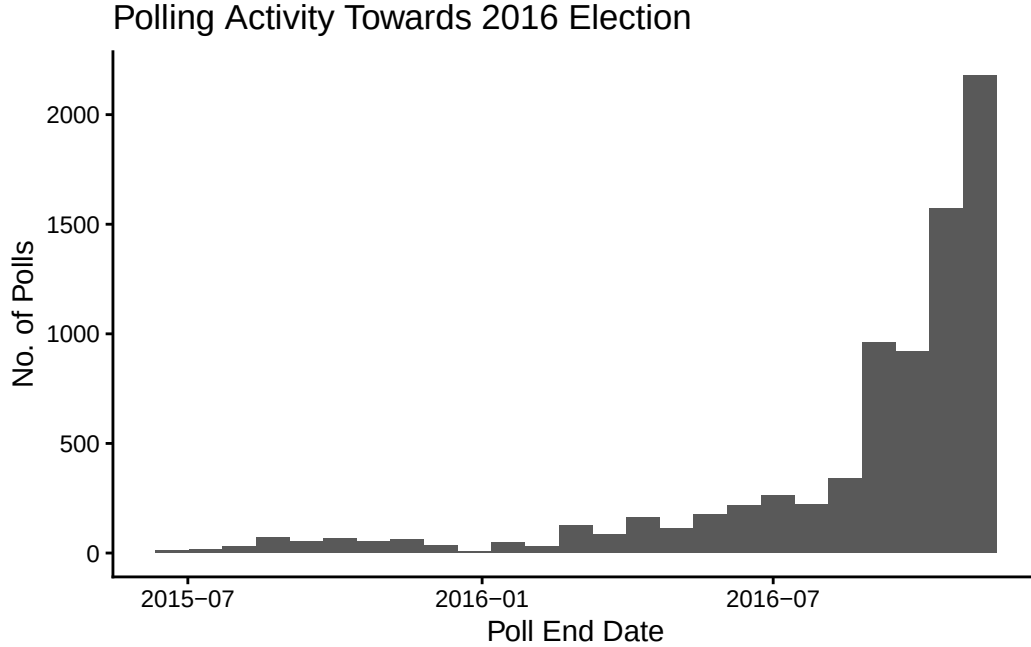


Figure 2: Polling coverage increased closer to Election Day in 2016, with more polls conducted in the last weeks of election campaign.

The number of observed polls are also not uniformly distributed throughout either 2012 or 2016 election cycle, with both data sets indicating greater polling activity towards Election Day. This means that polling observations in the later periods of campaigns are more heavily represented. This uneven temporal distribution should be considered when analysing summaries across the entire period.

Some candidates and response categories were also not included in every observation, leading to missing or non-applicable values. All together, such characteristics must be taken into consideration when comparing and interpreting polling results.

The 1976-2024 election data set consists of certified state-level returns for the US presidential elections. The purpose of this data set is to record realised election outcomes and thus, provides a foundation for comparison with polling estimates. It contains data from 1976 to 2024, retrieved from the MIT Election Data and Science Lab via Harvard Dataverse. Each observation records the votes for one candidate in a specific state and year. The data set fundamentally measures candidate vote counts and total number of state votes, along with information identifying the year of election, candidate, state and political party. For this particular analysis, the observations containing year 2012 and 2016 are specifically relevant given the availability of the polling data. The election data set 1976-2024-president.csv can be found in the data/raw/ folder within the [GitHub Repository](#)

Below is a detailed explanation of each column.

Key Variables:

- year: year which election was held
- state: name of state
- state_po: postal abbreviation of state
- state_fips, state_cen, state_ic: identification codes of state (FIPS code, Census state code, ICPSR state code)
- candidate: name of presidential candidate as it appears in House Clerk report
- party_detailed: detailed political party classification as they appear in House Clerk report
- party_simplified: simplified political party classification
- writein: if candidate was a write-in candidate
- candidatevotes: number of votes for the candidate for the particular party
- totalvotes: total votes in the state
- office: Office contested
- version: date the data set was last updated (format: yyyyymmdd)
- notes: additional information

Data Types:

- Numerical: candidatevotes, totalvotes
- Categorical*: state, state_po, party_detailed, party_simplified, office, candidate, state_fips, state_cen, state_ic
- Temporal: year
- Logical: writein
- Metadata: version, notes

*Last 3 variables are stored as numeric codes, however, treated as categorical given they identify states rather than represent quantities.

A notable limitation of the election data set is that results are aggregated and recorded only at the state level. This means that distinctions within a particular state (e.g. voting patterns between counties or urban/rural areas) cannot be investigated. Another key limitation for this analysis is the extent of comparison that can be made. Specifically, the election data set spans 1976 to 2024, while only 2 of these years overlap with the available polling data. This

limits the direct comparisons between polling predictions and actual elections results to 2012 and 2016 observations.

Description of Data Cleaning

All data sets were cleaned in R using the tidyverse package, while keeping a copy of the raw data set. They were cleaned to ensure variables were consistently represented while also handling invalid and missing values without removing any valid data. All data cleaning and wrangling is reproducible from the raw data sets using the scripts within the [GitHub Repository](#).

2012 polling data set

The poll_info variable was initially a single string with redundant information including identical election years and candidates across all observations. These were removed and renamed 'state' for meaningfulness.

The Obama and Romney variables indicated no missingness and thus, remained unchanged. Further, all values fell within the valid range of 0 to 100 percent, representing candidate support.

Missing values in the Undecided variable were unchanged as unreported values could neither be perceived to represent 0% undecided participants nor be accurately determined. All non-NA values fell within the valid range of 0 to 100 percent.

Values recorded as "Not included in poll" for the Other variable were changed to NA to ensure that missingness was represented consistently across variables in the data set. All non-NA values fell within the valid range of 0 to 100 percent.

Duplicate poll ID's were also identified with the poll_id column but were further examined before making changes. After investigating, it was noted that the repeated ID's did not correspond to any duplicated observations. Thus, all observations were retained.

Temporal variables start_date and end_date were converted from text to date objects to assist with potential temporal analysis such as examining candidate support over time. Further, it was confirmed that the start_date values was consistently earlier than the end_date values to ensure validity.

A validity check was also conducted on the sample_size variable, examining whether any values were unreasonable in the polling context. The sample_size values less than 1 were observed as invalid as there could not be less than 1 respondent in a poll. These values were recorded as NA rather than removing the observations, in order to retain other valid information.

A close look at the relationship between the partisan_affiliation and partisanship variable explained why many values in partisan_affiliation were recorded as "None".

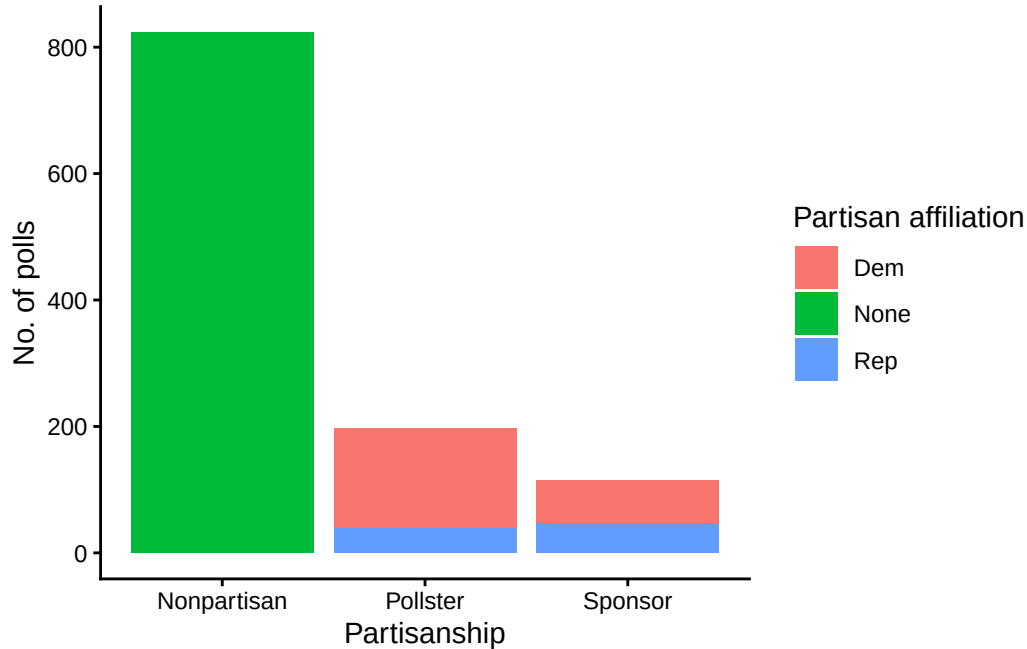


Figure 3: Missing partisan affiliation corresponds to non-partisan polls, indicating structural missingness.

The investigation indicated that these values were associated to polls that were listed “Non-partisan” in the partisanship column. Given that nonpartisan polls had no Republican or Democratic affiliation, these values indicate structural missingness rather than incomplete data collection. Therefore, observations were kept and nonpartisan polls were changed from “None” to NA.

Finally, categorical variables “pollster”, “sample_subpopulation” and “mode” were examined to check missingness. All variables were understood to contain no missingness.

The data set was transformed into a long format by reshaping the “Obama”, “Romney”, “Other”, and “Undecided” variables into the columns named “candidate” and “support”. Each observation (row) now represents the support for a particular candidate/response within a poll, allowing easier comparisons and visualisations of polling support. The cleaning script for the data set can be found in `src/clean_2012_polls.R`, with the cleaned data set, `state_polls_2012_clean.csv` in `data/clean/`.

2016 polling data set

The 2016 polling dataset was cleaned using the same methods as the 2012 dataset to ensure consistency comparison across years. The cleaning script for the data set can be found in `src/clean_2016_polls.r`, with the cleaned data set, `state_polls_2016_clean.csv` in `data/clean/`.

First, the “poll_info” variable was renamed to “state”, and all redundant information was removed to leave only state names. Any minor inconsistencies in state name formatting between the two datasets were standardized.

As with the 2012 data, the candidate support variables: “Trump”, “Clinton”, “Undecided”, “Other”, “Johnson”, and “McMullin”, were checked for missing values and validated to ensure percentages fell between 0 to 100%. The remaining variables (“poll_id”, “start_date”, “end_date”, “sample_size”, “partisan_affiliation”, “pollster”, “sample_subpopulation”, “mode”, and “Other”) were cleaned using the same workflow applied to the 2012 data set.

Finally, the candidate support columns were reshaped from wide to long format, creating two new variables: “candidate” and “support”. The consistency in structure ensured that the two polling results and election result could be analysed easily.

To explore the dataset’s structure, additional dataset visualization scripts were implemented into the src/ folder (e.g., src/clean_2016_polls_visualisations.R). These scripts analysed observations such as candidate margins across voter subpopulations (likely vs registered voters), weekly polling volume surges, undecided voter decay approaching election day, and state level margin distributions. These visualisations prompted the exploratory process, with the most interesting relationships being further examined in the EDA section.

1976-2024 election data set

The cleaning script for this data set can be found in src/clean_1976-2024_elections.R, with the cleaned data set, 1976-2024-president-clean.csv in data/clean/.

State names were standardised to title case to enable consistency with the two polling data sets. The office and notes variables were removed entirely, as office held an identical value for every observation and notes contained no recorded values at all as neither offered any useful variation for further analysis.

The version variable contained dates recorded in two different formats. These were standardised into a single consistent format and converted into a proper date object.

Candidate names were standardised by discarding doubled quotation marks, a formatting artifact introduced by inconsistent escaping in the raw file, which had otherwise caused the same candidate’s name to appear inconsistently across different rows. An examination of the year variable confirmed that all observations followed the expected four-year presidential election cycle, with no missing values identified.

The candidatevotes and totalvotes variables were examined for missingness and negative values. No missing values and no negative values were found in either variable, meaning all vote counts were considered valid.

The state identification variables state_po, state_fips, and state_cen indicated no missingness and thus remained unchanged. Missing state_ic codes were identified to correspond only to 2024 observations for New Hampshire and West Virginia. These values remained represented by NA, as the affected states were identifiable through the other state identifier variables.

The `party_simplified`, `party_detailed`, `write_in`, and candidate variables were investigated for missingness. Missing values represented by NA were retained to prevent any unsupported information from being introduced into the data set. Lastly, the cleaned data was examined for any entirely duplicated observations, with none identified.

Exploratory Data Analysis

We began our analysis by looking at national vote shares and state level winning trends from 1976 to 2024 via `src/clean_1976-2024_elections_visualisations.R`. The heatmap produced suggested that while many states remain consistent in their voting patterns, presidential elections usually come down to shifts in swing states.

Our investigation focuses on 2012 and 2016 presidential elections, since the polling data only covers these two periods.

Before comparing baseline candidate support against actual election results, a visualisation explored state by state average margins and their distributions using `src/clean_2016_polls_visualisations.r`. The produced bar chart and box plot suggested significant variance across states, prompting a further investigation into competitive states.

The following analysis explores how state election results transformed between 2012 and 2016, and further, whether differences in state competitiveness were observed in polling coverage and accuracy. For consistency, election and polling margins were calculated as Democratic - Republican (%). Positive values signify Democratic leaning observations whereas negative values signify Republican advantage.

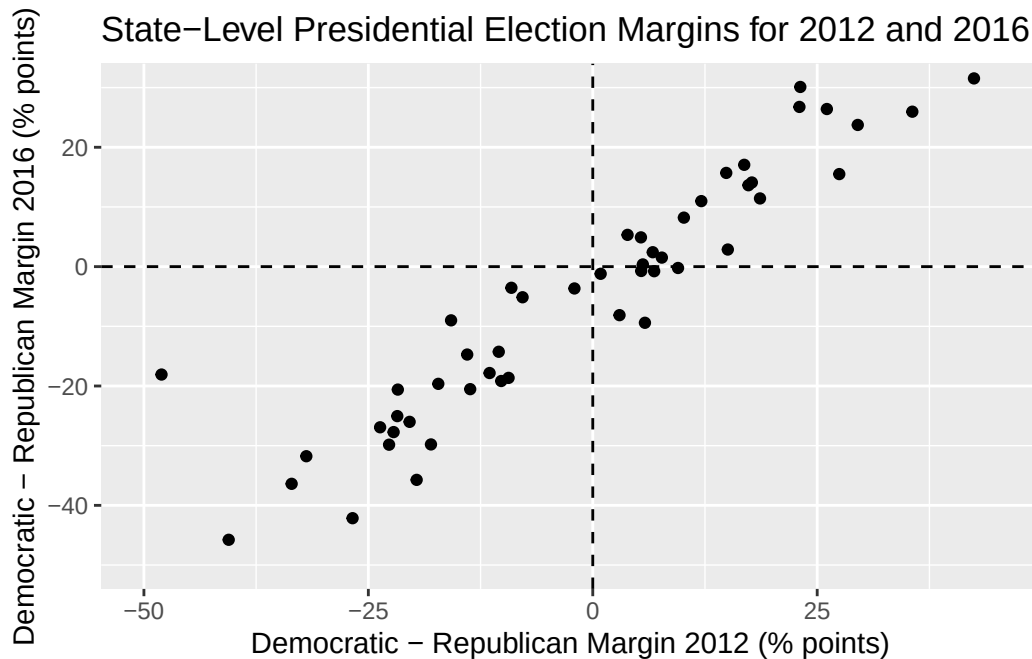


Figure 4: State election margins are mostly consistent in both years, but some states shifted from Democratic to Republican

The positive linear relationship indicates state election margins were relatively similar between 2012 and 2016, highlighting consistency in voting patterns by state. However, several states shifted from a Democratic in 2012 to Republican in 2016 as depicted in the bottom right quadrant. A number of these states were also close to a zero margin in 2016. This suggests that despite majority of US states remaining consistent in their overall voting, changes within competitive states were a key difference between the 2012 and 2016 elections.

The observation of competitive states led to an analysis on polling and how this was distributed across states. To determine state competitiveness, the absolute value of the election margin was measured, meaning that values near zero represented closer election results.

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`geom_smooth()` using formula = 'y ~ x'
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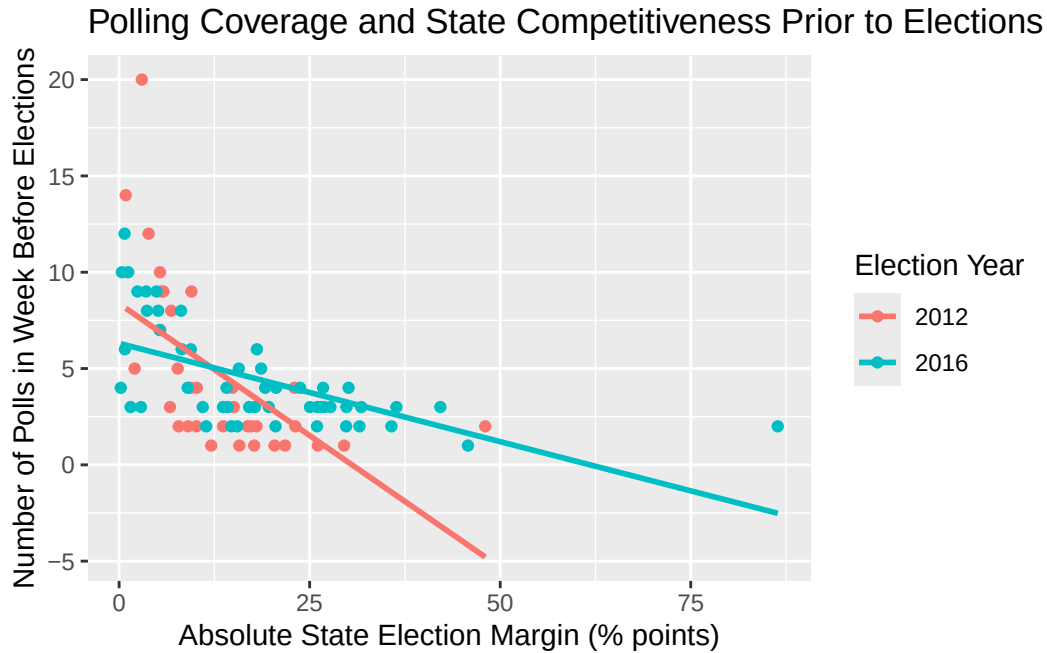


Figure 5: States with competitive election results appeared to receive greater polling coverage in week before elections for both 2012 and 2016

The negative linear relationship indicates that states with smaller election margins had mostly received more polls in the week prior to the elections, and vice versa. This implies that competitive states had more polling coverage rather than an even distribution across the states. It is important to note that the state competitiveness is measured utilising the election results, and thus, this relationship may be an association rather than a cause-and-effect evidence that shows competitiveness resulted in greater polling activity.

The differences in polling coverage led to an examination of the accuracy of the final week polling, and this was compared between the two elections. Polling error was measured as the polling margin - election margin. Positive errors indicated polls more Democratic than the election results, whereas, a negative error indicated the polls leaned more Republican.

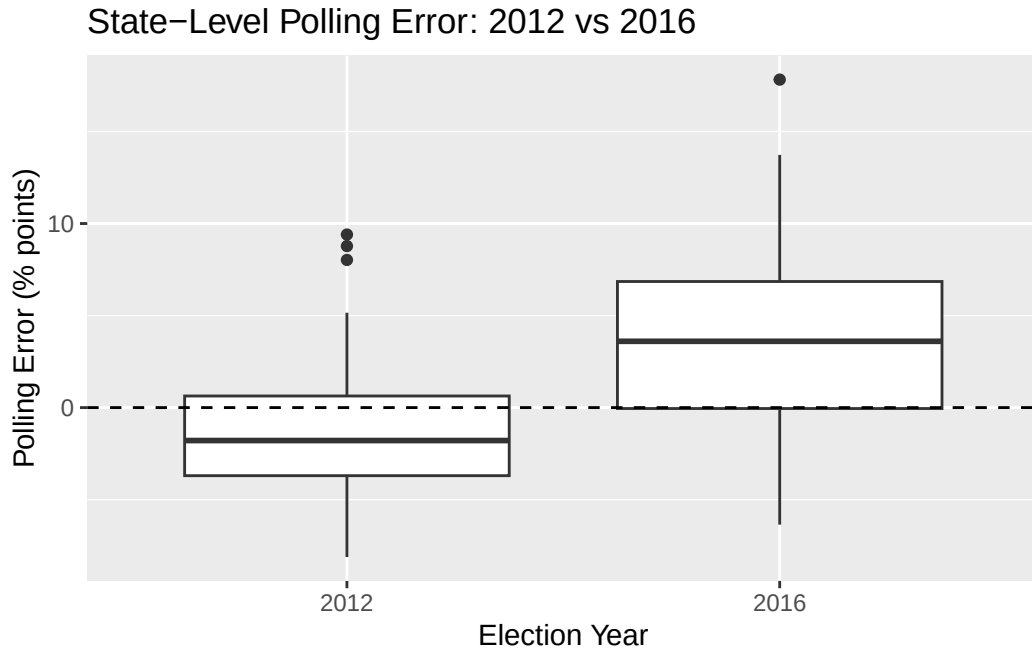


Figure 6: Polling leaning more Republican than actual election in 2012, whereas polling leaning more Democratic than actual election in 2016, with greater variation in 2016

While the 2012 polling error appears below zero, the 2016 error appears to be centred above zero and displays a greater spread. This difference corresponds to 2012 polls tending to be slightly more Republican than eventual results, and conversely, more Democratic in the 2016 polls. The greater variation in the 2016 polling error also indicates that polling margins were further from actual results than in comparison to 2012. An important limitation to consider is that the visualisation explores individual polls and with equal weight. This means that states with more polling coverage had greater influence than the states with fewer polls.

After observing a larger polling error for 2016, a closer examination on competitive states was conducted. Competitive states were chosen based on those that were within 10 percentage points for the election margin.

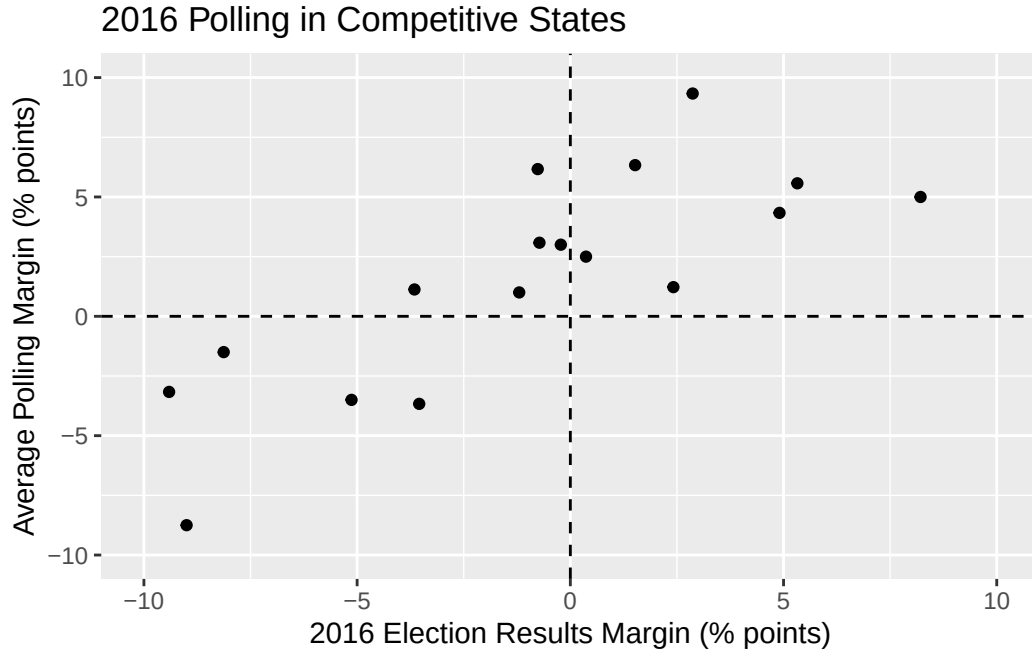


Figure 7: Many competitive states had average polling opposing the leading party from actual election votes

Majority of the ‘competitive’ states had a mean final week polling margin corresponding to the same side as the actual election results. However, some states had average polling margins contradicting the actual election results (top left quadrant). These particular states had a case where the polling margin indicated a different leading party from the party that eventually won the state. However, the averaging method means that each poll has equal weight and doesn’t consider differences in sample size, method of polling, or quality of pollster.

Additional state-level demographic/socioeconomic data including education, income, and urbanisation could further augment the presidential election results. This information could be useful in providing greater context on the changes in voting patterns between the two years, and also the differences in voting patterns across states. The additional data could be sourced via government agency websites such as U.S Census Bureau.

To further examine polling across time, it would be useful to collect additional data from other election years. This would allow analysis on polling accuracy and covered to be compared over time, enabling an investigation on whether the patterns observed in 2012 and 2016 signify a general pattern. The historical polling data would be available, but extent of coverage could range across elections.

Also, data on quality of pollsters and their differing methodologies (e.g. historical accuracy and method of sampling) could assist in understanding the differences in polling error, if looking

past the election year and competitiveness within states. This information would be available through polling organisations or external databases, but the extent could vary.

Research Questions

The exploratory data analysis observed that there was more polling coverage among competitive states, and also highlighted a greater variation in polling error in 2016. These findings prompted a deeper investigation into whether polling coverage results in greater accuracy. Previous research suggests that ‘pooling polls improves the precision of polling estimates’, but also indicates that polling methodologies can introduce biases that cannot be removed ‘if the biases work in the same direction’ (Sunshine, 2011). This motivates an examination on whether characteristics of polling are related to polling error. Below are two research questions and their potential ethical consequences.

Question 1: Does greater polling coverage enhance state-level polling accuracy, or do competitive states remain challenging to predict regardless of greater polling coverage?

Ethical consideration: A positive relationship between polling coverage and accuracy could lead to greater investments in polling, especially in more competitive states. Consequently, greater publications and poll attention could impact voter perceptions, turnout and allocation of resources for campaigns.

Question 2: To what extent are polling characteristics such as methodology, partisanship and sample size associated with greater state polling errors found in 2016 in comparison with 2012?

Ethical consideration: Associations between specific polling factors and errors must be communicated cautiously as this could unfairly impact public trust in particular pollsters/methodologies. This is important to consider as an observed association doesn’t directly mean that a certain method or pollster causes inaccurate data.

References

Sunshine, H D 2011, ‘The Evolution of Election Polling in the United States’, *Public Opinion Quarterly*, vol. 75, Oxford Academic, no. 5, pp. 962–981.